AWS Cloud-Based Web Server Development

This document explains the AWS Cloud-based web server development project in detail, covering setup, architecture, and key AWS services used.

Project Summary:

This project involves setting up a Windows-based web server using AWS EC2, and securing and optimizing it using other AWS services.

Key Components:

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1. EC2 (Elastic Compute Cloud) - Windows Server:
   - Hosted a static web application using IIS on a Windows EC2 instance.

2. CloudFront CDN:
   - Set up a CloudFront distribution for faster content delivery and high availability.
   - Linked to the EC2 public IP to cache and deliver content globally.

3. IAM Roles and Security Groups:
   - Created IAM roles to allow controlled access to services.
   - Configured security groups to allow only required ports (80, 3389).

4. CloudTrail Monitoring:
   - Enabled AWS CloudTrail to monitor and log all activities on the AWS account.

Setup Summary:

- Launch Windows EC2 instance
- Install IIS and host index.html file
- Create and connect CloudFront distribution
- Assign IAM role to instance
- Configure CloudTrail for logging

Result:

Successfully deployed a secure, fast, highly available cloud-based web server using AWS services.

Technologies Used:

- AWS EC2 (Windows Server)
- IIS Web Server
- Amazon CloudFront
- AWS IAM & Security Groups
- AWS CloudTrail

This project demonstrates cloud infrastructure setup, deployment, security, and monitoring in a real-world scenario.